COMPETENCY STANDARD

FOR

AUTO MECHANICS

(TRANSPORT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

This document is available at:

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:
Generic Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TRA-AM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks. 4. Respond to emergencies.	10
SICIP-TRA-AM-02-G	Carry out workplace interaction	1. Interpret workplace communication and etiquette. 2. Read and understand workplace documents. 3. Participate in workplace meetings and discussions. 4. Practice professional ethics at work.	10
SICIP-TRA-AM-03-G	Operate in a team environment	1. Identify team goals and work processes. 2. Identify own role and responsibilities within team. 3. Communicate and co-operate with team members. 4. Practice problem solving within the team.	10
		Total Hour	**30 Hrs.**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TRA-AM-01-S	Use hand and power tools	1. Identify and inspect hand and power tools. 2. Use hand tools properly and safely. 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools.	20
		Total Hour	**20 Hrs.**

OCCUPATION SPECIFIC COMPETENCIES (310 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-TRA-AM-01-O	Understand automotive engine	1. Identify types of engines. 2. Identify major components of engine and other systems. 3. Identify major components of VVTi engine	20
SICIP-TRA-AM-02-O	Perform engine overhauling	1. Identify types of engine overhauling. 2. Check the necessity of overhauling. 3. Carry out engine top overhauling. 4. Carry out engine minor overhauling. 5. Carry out engine major overhauling.	70
SICIP-TRA-AM-03-O	Service auxiliary systems	1. Service starting system. 2. Service fuel system. 3. Service cooling system. 4. Service lubricating system. 5. Service ignition system. 6. Service electrical and electronic system.	50
SICIP-TRA-AM-04-O	Service power transmission system	1. Service clutch system. 2. Service gear box. 3. Service differential system. 4. Service propeller shaft and all related components. 5. Service all sensors	40
SICIP-TRA-AM-05-O	Service control system	1. Service brake system. 2. Service steering system.	30
SICIP-TRA-AM-06-O	Service suspension system	1. Test and service shock absorber. 2. Test and service springs. 3. Test and service torsion and stabiliser bar. 4. Test and service bush and mountings.	40
SICIP-TRA-AM-07-O	Service hybrid system	1. Understand hybrid system components 2. Diagnose hybrid system faults 3. Service hybrid batteries 4. Maintain electric motor and inverter systems 5. Troubleshoot hybrid system issues	60

Code	Unit of Competency	Elements of Competency	Duration (hours)
		Total Hours	310 Hrs.

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The Generic Competencies

Unit of Competency:	Nominal Duration:	Unit Code:
APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	10 Hrs**.**	SICIP-TRA-AM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practice in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are interpreted. 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are applied in the workplace including **<u>personal protective equipment (PPE)</u>**. 2.2 Common health issues are recognised. 2.3 Common safety issues are identified.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted.
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 **<u>Emergency response plans and procedures</u>** are responded to. 4.3 **<u>First aid procedures</u>** during emergency situations are identified.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. OHS policies	1.1 Organisational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Emergency response plans and procedures	2.1 Firefighting procedures 2.2 Earthquake response procedures 2.3 Emergency response plans and procedures 2.4 Medical and first aid

3. First aid procedure	3.1 Washing of open wound 3.2 Washing chemically infected area 3.3 Applying bandage 3.4 Taking appropriate medicine
4. Personal protective equipment	4.1 Safety glasses 4.2 Ear plugs 4.3 Gloves 4.4 Apron 4.5 Helmet 4.6 Mask 4.7 Safety shoes

Curricular Content Guide

1. Underpinning Knowledge	1.1. Workplace OHS policies and procedures 1.2. Work safety procedures 1.3. Emergency response procedures: 1.3.1. Firefighting 1.3.2. Earthquake response 1.3.3. Accident response 1.4. Types of hazards (biological, chemical and physical) and their effects 1.5. OHS awareness 1.6. Personal protective equipment (PPE)
2. Underpinning Skills	2.1. Identifying OHS policies and procedures 2.2. Applying personal health and safety practices 2.3. Reporting hazards and risks 2.4. Responding to emergencies
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Projector 4.7 Stationary

	4.8 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT WORKPLACE INTERACTION**	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TRA-AM-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions, and practicing professional ethics at work.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organisational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **courteous manner** to gather and convey information. 1.4 **Workplace procedures and matters** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **appropriate sources**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning Knowledge	1.1. Workplace communication and etiquette 1.2. Workplace documents, signs and symbols 1.3. Meeting procedure and etiquette 1.4. Professional ethics
2. Underpinning Skills	2.1. Demonstrating workplace communication and etiquette 2.2. Interpreting workplace instructions and symbols 2.3. Demonstrating active participation in workplace meeting 2.4. Applying professional ethics at work
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures

	4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TRA-AM-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and co-operating with team members, and practicing problem solving within the team.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in <u>bold and underlined</u> are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Roles and objectives of the team are identified and interpreted. 1.2 Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and co-operate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through **<u>sharing information</u>** or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognised and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Sharing information	1.1 Agenda 1.2 Minutes 1.3 Progress and incident reports 1.4 Operational manuals

	1.5 Visual and graphic materials 1.6 Emails and SMS 1.7 Phone directory 1.8 Policy, procedure and standards 1.9 OHS information

Curricular Content Guide

1. Underpinning Knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities 1.3. Finding problems and solving them
2. Underpinning Skills	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Projector 4.3. Stationary 4.4. Learning manual 4.5. Tools, equipment and facilities appropriate to the process or activities.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted team goals and work processes 1.2 Described roles and responsibilities of team members 1.3 Communicated and co-operated with team members 1.4 Demonstrated problem solving within the team
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector-specific Competencies

Unit of Competency: **USE HAND AND POWER TOOLS**	**Nominal Duration:** 20 Hrs.	**Unit Code:** SICIP-TRA-AM-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to use hand and power tools in the workplace. It specifically includes the tasks of identifying and inspecting hand and power tools for usability, using and operating tools properly and safely, and cleaning and maintaining hand and power tools after use.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognised. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate **hand tools** are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 **Measuring tools** are checked and calibrated before use. 2.5 Use hand and measuring tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1 Appropriate **power tools** are selected. 3.2 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3 Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4 Proper sequence of operation applied for using power tools. 3.5 Unsafe or faulty power tools are identified and marked for repair. 3.6 Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1 Appropriate **power tools** are selected. 4.2 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements.

	4.3 Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 4.4 Proper sequence of operation applied for using power tools. 4.5 Unsafe or faulty power tools are identified and marked for repair. 4.6 Operate power tools properly and safely to perform work activity.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Hand tools	1.1. Hacksaw 1.2. Hammer 1.3. Files 1.4. Pliers 1.5. Punches 1.6. Screwdrivers 1.7. Wrench box 1.8. Hand tap 1.9. Wire cutters 1.10. Hand hacksaw 1.11. Drill 1.12. Grinder 1.13. Dial gauge 1.14. Spanner comb 1.15. Spanner ring 1.16. Socket ratchet set 1.17. Easy opener 1.18. Top roller adjust gauge 1.19. Allen key 1.20. Top roller adjust gauge
2. Power tools	2.1 Portable drilling machine 2.2 Threading machine 2.3 Saws 2.4 Glue gun 2.5 Soldering iron 2.6 Grinders
3. Measuring tools	3.1 Gauges 3.2 Hose level 3.3 Water level 3.4 Micrometre 3.5 Multi-meter 3.6 Tachometer 3.7 Thermometers 3.8 Dial indicators 3.9 Slide callipers

	3.10 Tape measure 3.11 Testing lead

Curricular Content Guide

1. Underpinning Knowledge	1.1. Information on types of hand and power tools, their functions and use 1.2. Procedures for safely using hand and power tools
2. Underpinning Skills	2.1. Identifying hand, power and measuring tools 2.2. Following safety precautions when using hand, power and measuring tools 2.3. Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4. Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5. Cleaning and maintaining hand and power tools after use 2.6. Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Hand tools 4.3 Power tools 4.4 Measuring tools 4.5 Projector 4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified and selected appropriate hand and power tools for work to be performed 1.2 Identified and used measuring and testing tools appropriate to work activity

	1.3 Followed safety precautions when using hand and power tools 1.4 Operated power tools safely and pursuant to manufacturer's operating specification 1.5 Performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: **UNDERSTAND AUTOMOTIVE ENGINE**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-TRA-AM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand automotive engine. It specifically includes the tasks of identifying types of engines, identifying major components of engine and other systems, and identifying major components of VVTi engine.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify types of engines	1.1. Different **types of internal combustion (IC) engines** are identified and understood. 1.2. Working principles of engine are explained. 1.3. Engine classifications are recognized.
2. Identify major components of engine and other systems	2.1 **Major components of engine** are identified. 2.2 **Key components of the fuel system**, are identified and their functions explained. 2.3 Importance of the **cooling system components** are understood. 2.4 **Ignition system components** are identified and explained. 2.5 **lubrication system component** is recognized and understood. 2.6 **Exhaust system components** are identified and understood.
3. Identify major components of VVTi engine	3.1 **Major components of VVTi engine** are identified. 3.2 Major components of VVTi engine are described.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of internal combustion (IC) engines	1.1. Based on fuel 1.1.1. Gasoline (petrol/octane/gas) 1.1.2. Diesel 1.2. Based on operating system 1.2.1. Carburettor 1.2.2. EFI 1.2.3. VVTi 1.2.4. Hybrid
2. Major components of engine	2.1 Engine head 2.2 Engine block

	2.3 Oil pan 2.4 Piston 2.5 Connecting rod 2.6 Crankshaft 2.7 Camshaft 2.8 Engine valves 2.9 Fly wheel 2.10 Bearing
3. Key components of the fuel system	3.1 Fuel injector (diesel and petrol) 3.2 Carburetor 3.3 Fuel pump (mechanical and electrical) 3.4 High pressure fuel pump 3.5 Fuel filter 3.6 Fuel pressure regulator 3.7 Fuel gallery 3.8 Electronic control unit (ECU)
4. Cooling system components	4.1 Radiator 4.2 Thermostat valve 4.3 Water pump 4.4 Cooling fans 4.5 Pressure cap 4.6 Water temperature sensor 4.7 Electronic control unit (ECU)
5. Ignition system components	5.1 Spark plugs 5.2 Ignition coil (conventional and electronic) 5.3 Distributor
6. Lubrication system components	6.1 Oil pump 6.2 Oil filter 6.3 Oil pan 6.4 Oil pressure switch 6.5 Oil pressure relief valve
7. Exhaust system components	7.1 Exhaust manifold 7.2 Catalytic converter 7.3 Oxygen sensor 7.4 Muffler 7.5 Tailpipe
8. Major components of VVTi engine	8.1 VVT-i Solenoid 8.2 Oil Control Valve 8.3 Ignition coil 8.4 VVTi pulley

Curricular Content Guide

1. Underpinning Knowledge	1.1. Internal combustion (IC) engine and its types 1.2. Working principles of engine 1.3. Major components of engine 1.4. Key components of the fuel system 1.5. Importance of the cooling system components 1.6. Ignition system components

	1.7. lubrication system component 1.8. Exhaust system components 1.9. Major components of VVTi engine
2. Underpinning Skills	2.1. Identifying different types of internal combustion (IC) engines 2.2. identifying key components of fuel system 2.3. Identifying Ignition system components 2.4. Recognizing lubrication system component 2.5. Identifying Exhaust system components 2.6. Identifying Major components of VVTi engine
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of engines. 1.2 identified major components of engine and other systems. 1.3 identified major components of VVTi engine
2 Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM ENGINE OVERHAULING**	Nominal Duration: 70 Hrs.	Unit Code: SICIP-TRA-AM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform engine overhauling. It specifically includes the tasks of identifying types of engine overhauling, checking the necessity of overhauling, carrying out engine top overhauling, carrying out engine minor overhauling and carrying out engine major overhauling.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of engine overhauling	1.1 **Different types of engine overhauling** are identified and understood. 1.2 Process of top overhauling is explained. 1.3 Process of minor overhauling is identified and understood. 1.4 Major overhauling procedures are explained.
2. Check the necessity of overhauling	2.1 Engine's performance is assessed to determine if overhauling is necessary. 2.2 **Diagnostic tests** are conducted to evaluate the condition of engine components. 2.3 **Condition of the major engine components** is checked for wear, cracks, or damage. 2.4 Engine's oil consumption, exhaust emissions, and coolant system performance are monitored. 2.5 Presence of metal particles or contaminants in the oil filter is checked.
3. Carry out engine top overhauling	3.1 Engine is prepared for top overhauling by disconnecting necessary components. 3.2 Engine is disassembled and inspected for damage or wear. 3.3 Cylinder head is cleaned and necessary repairs are carried out. 3.4 Related components are inspected for damage. 3.5 Piston rings and other related components are inspected. 3.6 Gasket surfaces are cleaned and prepared for new gaskets. 3.7 Cylinder head is re-installed, and torque specifications are followed during re-assembly. 3.8 Engine is re-assembled, and the timing system is checked and adjusted for proper operation.
4. Carry out engine minor overhauling	4.1 Engine is prepared for minor overhauling by disconnecting necessary components.

	4.2 Components are inspected and checked. 4.3 Cylinder head is inspected and checked for cracks, leaks, or damage. 4.4 Valve components are checked and any faulty components are replaced. 4.5 Timing system is inspected and adjusted to ensure proper operation.
5. Carry out engine major overhauling	5.1 Engine is prepared and disconnected from all necessary **systems**. 5.2 Gear box and all mountings are disassembled from engine. 5.3 Engine is securely detached from the chassis according to safety protocols. 5.4 Camshaft and crankshaft are inspected and replaced as necessary. 5.5 All necessary engine repair kits are replaced in accordance with manufacturer specifications. 5.6 All components are reassembled following the manufacturer's specifications. 5.7 Timing system is inspected and replaced following to factory specifications. 5.8 Engine is tested after reassembly, ensuring that it meets performance standards.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of engine overhauling	1.1 Top overhauling 1.2 Minor overhauling 1.3 Major overhauling
2. Diagnostic tests	2.1 Compression test 2.2 Leakage test 2.3 Oil analysis 2.4 Smoke test
3. Condition of the major engine components	3.1 Pistons 3.2 Cylinders 3.3 Valves 3.4 Bearings 3.5 Thrust washer
4. System	4.1 Battery 4.2 Fuel system 4.3 Electrical connections 4.4 Intake system 4.5 Exhaust system 4.6 Starting system
5. Valve system	5.1 Camshaft 5.2 Valves 5.3 Valve springs 5.4 Lifters

Curricular Content Guide

1. Underpinning Knowledge	1.1. Engine overhauling and its types 1.2. Process of top overhauling 1.3. Process of minor overhauling 1.4. Major overhauling procedures 1.5. Engine's oil consumption, exhaust emissions and coolant system
2. Underpinning Skills	2.1 Identifying types of engine overhauling 2.2 Conducting diagnostic tests to evaluate condition of engine components 2.3 Checking condition of the major engine components for wear, cracks or damage 2.4 Monitoring engine's oil consumption, exhaust emissions and coolant system performance 2.5 Checking presence of metal particles or contaminants in the oil filter 2.6 Disassembling and inspecting engine and related components for damage or wear (for engine top overhauling) 2.7 Carrying out necessary repairs and cleaning of engine and related components (for engine top overhauling) 2.8 Re-assembling engine and related components (for engine top overhauling) 2.9 Disconnecting necessary components, and inspecting and checking the same (for engine minor overhauling) 2.10 Re-connecting the components after necessary repairs and cleaning of the components (for engine minor overhauling) 2.11 Disassembling gear box and all mountings from engine, detaching engine from the chassis and disconnecting other related components (for engine major overhauling) 2.12 Re-assembling the engine and related components after necessary repairs of the components (for engine major overhauling)
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of engine overhauling. 1.2 checked the necessity of overhauling. 1.3 carried out engine top overhauling. 1.4 carried out engine minor overhauling. 1.5 carried out engine major overhauling.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE AUXILIARY SYSTEMS**	Nominal Duration: 50 Hrs.	Unit Code: SICIP-THS-AM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform services auxiliary systems. It specifically includes the tasks of servicing starting system, servicing fuel system, Servicing cooling system, Servicing lubricating system, Servicing ignition system, and Servicing electrical and electronic system.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Service starting system	1.1 **Major components of starting system** are identified. 1.2 Functions of major components are described. 1.3 Major components are tested and replaced, if necessary.
2. Service fuel system	2.1 **Major components of fuel system** are identified. 2.2 Functions of major components are described. 2.3 Major components are tested and replaced, if necessary.
3. Service cooling system	3.1 **Major components of cooling system** are identified. 3.2 Functions of major components are described. 3.3 Fan belt tension is adjusted. 3.4 Radiator and engine flushing is performed. 3.5 Major components are tested and replaced, if necessary.
4. Service lubricating system	4.1 **Major components of lubrication** system are identified. 4.2 Functions of major components are described. 4.3 Major components are tested and replaced, if necessary
5. Service ignition system	5.1 **Major components of ignition** system are identified. 5.2 Functions of major components are described. 5.3 Major components are tested and replaced, if necessary.
6. Service electrical and electronics system	6.1 Relay, fuse, and fuse box are inspected 6.2 Any faulty components are replaced or repaired to ensure proper function. 6.3 ECU (Engine Control Unit)/ECM (Engine Control Module) and BCM (Body Control Module) are inspected and replaced if necessary. 6.4 Transistors, capacitors/condenser, and ICs (integrated circuits) are tested.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Major components of starting system	1.1 Starting motor 1.2 Solenoid switch 1.3 Ignition switch 1.4 Starting relay

	1.5 Battery
2. Major components of fuel system	2.1 Fuel pump 2.2 Fuel filter 2.3 Petrol Injector 2.4 Fuel pressure regulator
3. Major components of cooling system	3.1 Water pump 3.2 Radiator with pressure cap 3.3 Cooling fan with motor and thermostatic switch 3.4 Thermostat valve 3.5 Hose pipes 3.6 Water temperature sensor
4. Major components of lubrication	4.1 Lubricating oil pump 4.2 Oil filter 4.3 Pressure relief valve 4.4 Oil pressure switch
5. Major components of ignition system	5.1 Distributor 5.2 Ignition coil 5.3 Spark plug 5.4 Ignitor 5.5 Battery

Curricular Content Guide

1. Underpinning Knowledge	1.1. Functions of major components of starting system 1.2. Functions of major components of fuel system 1.3. Functions of major components of cooling system 1.4. Functions of major components of lubrication system 1.5. Functions of major components of ignition system
2. Underpinning Skills	2.1. Identifying major components of starting system 2.2. Identifying major components of fuel system 2.3. Identifying major components of cooling system 2.4. Identifying major components of lubrication system 2.5. Identifying major components of ignition system 2.6. Testing and replacing, if necessary, major components of starting, fuel, cooling, lubrication and ignition systems 2.7. Inspecting relay, fuse and fuse box, and replacing or repairing faulty components to ensure proper functioning 2.8. Inspecting ECU (Engine Control Unit)/ECM (Engine Control Module) and BCM (Body Control Module), and replacing if necessary 2.9. Testing transistors, capacitors/condenser, and ICs (integrated circuits)
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties

	3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 serviced starting system. 1.2 serviced fuel system. 1.3 serviced cooling system. 1.4 serviced lubricating system. 1.5 serviced ignition system. 1.6 serviced electrical and electronic system.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE POWER TRANSMISSION SYSTEM**	Nominal Duration: 40 Hrs**.**	Unit Code: SICIP-TRA-AM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform service power transmission system. It specifically includes the tasks of servicing clutch system, servicing gear box, servicing differential system, servicing propeller shaft & all related components, and servicing all sensors.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Service clutch system	1.1 **Major components of clutch system** are identified. 1.2 Functions of major components are described. 1.3 Adjustment and bleeding of clutch system is carried out. 1.4 Major components are tested and replaced, if necessary.
2. Service gear box	2.1 **Major components of gear box** are identified. 2.2 Functions of major components are described. 2.3 Change of gear oil is carried out. 2.4 Major components are tested and replaced, if necessary.
3. Service differential gear box	3.1 **Major components of differential gear box** are identified. 3.2 Functions of major components are described. 3.3 Adjustment and oil change is carried out. 3.4 Major components are tested and replaced, if necessary
4. Service propeller shaft and all related components	4.1 **Major components of propeller shaft and universal joint** are identified. 4.2 Functions of major components are described. 4.3 Major components are tested and replaced, if necessary.
5. Service all sensors	5.1 **Major sensors** of car and engine is identified. 5.2 Sensor faults are checked as required. 5.3 Different types of sensors are serviced and replaced. 5.4 Sensors are tested to ensure proper functioning.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Major components of clutch system	1.1 Clutch plate/friction plate 1.2 Pressure plate 1.3 Coil spring/diaphragm spring 1.4 Release bearing 1.5 Slave cylinder
2. Major components of gear box	2.1 Input shaft/clutch shaft 2.2 Lay shaft/counter shaft

	2.3 Main shaft/output shaft 2.4 Synchronizing unit 2.5 Different gears
3. Major components of differential gear box	3.1 Crown wheel 3.2 Sun gear 3.3 Star pinion 3.4 Pinion shaft 3.5 Casing
4. Major components of propeller shaft and universal joint	4.1 Propeller shaft 4.2 Universal joint 4.3 Slip joint
5. Major sensors	5.1 Crank position 5.2 Cam position 5.3 Water temperature 5.4 Air temperature 5.5 Throttle position 5.6 MAP 5.7 MAF 5.8 Oxygen 5.9 Knock 5.10 Idle speed

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions of major components of clutch system 1.2 Functions of major components of gear box 1.3 Functions of major components of differential gear box 1.4 Functions of major components of propeller shaft and universal joint 1.5 Different types of sensors
2. Underpinning Skills	2.1 Identifying major components of clutch system 2.2 Identifying major components of gear box 2.3 Identifying major components of differential gear box 2.4 Identifying major components of propeller shaft and universal joint 2.5 Identifying major sensors of car and engine 2.6 Checking sensor faults as required 2.7 Servicing and replacing different types of sensors 2.8 Testing sensors to ensure proper functioning 2.9 Testing and replacing, where necessary, major components of clutch, gear box, differential gear box, propeller shaft and universal joint 2.10 Carrying out adjustment and bleeding of clutch system, change of gear oil, Adjustment and oil change
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety

	3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Facilities appropriate to the process/activities 4.5 Materials relevant to the proposed activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 serviced clutch system. 1.2 serviced gear box. 1.3 serviced differential system. 1.4 serviced propeller shaft and all related components. 1.5 serviced all sensors
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE CONTROL SYSTEM**	**Nominal Duration:** 30 Hrs**.**	**Unit Code:** SICIP-TRA-AM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform service control system. It specifically includes the tasks of servicing brake system and steering system.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in <u>bold and underlined</u> are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Service brake system	1.1 <u>**Major components of brake system**</u> are identified. 1.2 Functions of major components are described. 1.3 Functions of anti-lock brake system (ABS) is described. 1.4 Adjustments and bleeding of brake system is carried out. 1.5 Major components are tested and replaced, if necessary
2. Service steering system	2.1 <u>**Major components of steering system**</u> are identified. 2.2 Functions of major components are described. 2.3 Adjustment of steering system is carried out. 2.4 Major components are tested and replaced, if necessary

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Major components of brake system	1.1 Brake master cylinder 1.2 Servo-cylinder/brake buster 1.3 Wheel cylinder 1.4 Brake shoe assembles 1.5 Brake adjuster 1.6 Brake drum 1.7 Brake pad and calliper
2. Major components of steering system	2.1 Steering wheel with column 2.2 Steering gear box 2.3 Tie rod with tie rod end 2.4 Steering pump 2.5 Steering motor

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions of major components of brake system 1.2 Functions of anti-lock brake system (ABS) 1.3 Functions of major components of steering system
2. Underpinning Skills	2.1 Identifying major components of brake system 2.2 Carrying out adjustments and bleeding of brake system 2.3 Testing and replacing major components of brake system and ABS (if necessary) 2.4 Identifying Major components of steering system

	2.5 Carrying out adjustment of steering system 2.6 Testing and replacing major components of steering system (if necessary)
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Measuring and testing instruments

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 serviced brake system. 1.2 serviced steering system.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE SUSPENSION SYSTEM**	**Nominal Duration:** 40 Hrs**.**	**Unit Code:** SICIP-TRA-AM-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform service suspension system. It specifically includes the tasks of testing and servicing shock absorber, testing and servicing springs, testing and servicing torsion and stabilizer bar, and testing and servicing bush and mountings.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Test and service shock absorber	1.1 **Types of shock absorber** are identified. 1.2 Function of shock absorber are described. 1.3 Shock absorber is tested and replaced, if necessary
2. Test and service springs	2.1 **Components of leaf spring** are identified. 2.2 Types of leaf spring and coil spring are identified 2.3 Function of leaf spring and coil spring are described. 2.4 Service of leaf spring is carried out. 2.5 Leaf and coil spring are tested and replaced, if necessary.
3. Test and service torsion and stabiliser bar	3.1 Function of torsion and stabiliser bar described. 3.2 Service of torsion bar and stabiliser bar is carried out. 3.3 Torsion and stabiliser bar are tested and replaced, if necessary.
4. Test and service bush and mountings	4.1 **Type of bush and mountings** are identified. 4.2 Functions of bush and mountings are described. 4.3 Bush and mountings are tested and replaced, if necessary.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of shock absorber	1.1 Air type 1.2 Oil type
2. Components of leaf spring	2.1 Leaves of leaf spring 2.2 Centre bolt and U-bolt 2.3 Shackle bar and hanger
3. Types of bush and mountings	3.1 Shock absorber 3.2 Torsion bar 3.3 Stabiliser bar 3.4 Engine mounting 3.5 Gear box mounting 3.6 Main mounting

Curricular Content Guide

1. Underpinning Knowledge	1.1. Functions of shock absorber 1.2. Types and functions of leaf spring and coil spring 1.3. Functions of torsion and stabiliser bar 1.4. Types and functions of bush and mountings
2. Underpinning Skills	2.1. Identifying types of shock absorber 2.2. Identifying components of leaf spring 2.3. Identifying types of leaf spring and coil spring 2.4. Carrying out service of leaf spring 2.5. Carrying out service of torsion bar and stabiliser bar 2.6. Identifying type of bush and mountings 2.7. Testing and replacing shock absorber, leaf and coil spring, torsion and stabiliser bar, and bush and mountings, if necessary
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Measuring and testing instruments 4.5 Drawings, specifications, and manuals 4.6 Replacement products 4.7 Projector 4.8 Stationary 4.9 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 tested and serviced shock absorber. 1.2 tested and serviced springs. 1.3 tested and serviced torsion and stabiliser bar. 1.4 tested and serviced bush and mountings.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning

	2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE HYBRID SYSTEM**	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TRA-AM-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to service hybrid system. It specifically includes the tasks of understanding hybrid system components, diagnosing hybrid system faults, servicing hybrid batteries, maintaining electric motor & inverter systems, and troubleshooting hybrid system issues.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand Hybrid System Components	1.1. **Key components of a hybrid system** are understood and identified. 1.2. **Types of hybrids system** are identified. 1.3. **Benefits of hybrid systems** are described. 1.4. **Functions of the electric motor** are recognized and explained. 1.5. **Functions of the internal combustion engine (ICE) in hybrid systems** is explained. 1.6. **Role of the transmission system** in hybrid vehicles is recognized. 1.7. Integration of hybrid components is understood to improve fuel efficiency and reduce emissions. 1.8. Basic operating principles of a hybrid system is understood.
2. Diagnose Hybrid System Faults	2.1 **Hybrid system faults** are identified through diagnostic tools and techniques. 2.2 Error codes and diagnostic trouble codes (DTCs) are retrieved from the ECU (Engine Control Unit). 2.3 Performance of the electric motor, battery pack, and power electronics is assessed. 2.4 **Battery** health is diagnosed by measuring charge levels, voltage, and state of charge (SOC). 2.5 Faults in the internal combustion engine (ICE) are diagnosed through performance tests and mechanical inspections. 2.6 Transmission system is checked for faults in hybrid-specific components.
3. Service Hybrid Batteries	3.1 Hybrid battery is inspected and their condition assessed. 3.2 Battery charge level and voltage balance across cells are checked using diagnostic tools to assess battery condition. 3.3 Battery terminals and connections are inspected for corrosion, and any buildup is cleaned to ensure proper

	electrical contact. 3.4 Battery cooling system is checked for proper operation. 3.5 Battery management system (BMS) is tested. 3.6 Faulty or damaged battery cells are identified and replaced as necessary. 3.7 Battery is recharged if necessary, following the manufacturer's guidelines.
4. Maintain Electric Motor and Inverter Systems	4.1 Electric motor and inverter system are inspected. 4.2 Cooling system of the electric motor and inverter is checked. 4.3 Electrical connections and wiring in the motor and inverter systems are inspected. 4.4 Inverter system is tested to ensure it is converting DC power from the battery to AC power for the electric motor efficiently. 4.5 Motor brushes are inspected for wear and replaced if necessary. 4.6 Power electronics associated with the inverter are checked. 4.7 Diagnostic tools are used to check for any error codes. 4.8 Electric motor and inverter system are tested after maintenance.
5. Troubleshoot Hybrid System Issues	5.1 Hybrid system diagnostics are performed using appropriate tools and software. 5.2 Error codes retrieved from the ECU (Engine Control Unit) or hybrid control module are analyzed. 5.3 Electric motor, battery pack, power electronics, and internal combustion engine (ICE) are inspected. 5.4 Battery condition is checked by measuring voltage levels, state of charge (SOC), and temperature. 5.5 Power electronics system is tested. 5.6 Wiring and electrical connections are inspected. 5.7 Transmission system is examined. 5.8 Regenerative braking functionality is tested. 5.9 Hybrid system control modules are checked.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Key components of a hybrid system	1.1 Internal combustion engine (ICE) 1.2 Electric motor 1.3 Battery pack 1.4 Power electronics 1.5 Transmission
2. Types of hybrids system	2.1 Series Hybrid 2.2 Parallel Hybrid 2.3 Plug-in Hybrid 2.4 Mild-Hybrid
3. Benefits of hybrid	3.1 Fuel efficiency

systems	3.2 Reduced emissions 3.3 Re-generative braking 3.4 Smooth performance
4. Functions of the electric motor	4.1 Primary Power Source at Low Speeds 4.2 Regenerative Braking 4.3 Assist to the Internal Combustion Engine (ICE) 4.4 Battery Charging 4.5 Silent Operation
5. Functions of the internal combustion engine (ICE) in hybrid systems	5.1 Primary Power Source (provides the primary power for the vehicle) 5.2 Battery Charging 5.3 Supplementing Electric Motor 5.4 Regenerative Braking Integration
6. Role of the transmission system	6.1 Power Distribution Between ICE and Electric Motor 6.2 Smooth Transition Between Power Sources 6.3 Regenerative Braking Integration 6.4 Optimizing Efficiency 6.5 Energy Flow Management 6.6 Enhancing Power Delivery in Complex Hybrid Systems 6.7 Transmission Efficiency in Different Drive Modes 6.8 Managing Overdrive and Torque Distribution
7. Hybrid system faults	7.1 Internal combustion engine (ICE) 7.2 Electric motor 7.3 Battery pack 7.4 Power electronics 7.5 Transmission
8. Battery	8.1 High-voltage lithium-ion (Li-ion) batteries 8.2 Lead-acid batteries

Curricular Content Guide

1. Underpinning Knowledge	1.1. Key components and benefits of hybrid systems 1.2. Functions of the electric motor, internal combustion engine (ICE) and role of the transmission system in hybrid system/vehicle 1.3. Basic operating principles of a hybrid system 1.4. Functions of the electric motor, and internal combustion engine (ICE) in hybrid systems 1.5. Role of the transmission system in hybrid vehicles 1.6. Hybrid battery and its types 1.7. Hybrid system control modules
2. Underpinning Skills	2.1. Identifying Types of hybrid system and key components of a hybrid system 2.2. Identifying hybrid system faults 2.3. Retrieving error codes and diagnostic trouble codes (DTCs) from ECU (Engine Control Unit) 2.4. Assessing performance of electric motor, battery pack and power electronics 2.5. Diagnosing/checking battery health, faults in the ICE and

	transmission system for faults in hybrid-specific components 2.6. Checking/testing battery cooling system, battery management system (BMS) and faulty or damaged battery cells 2.7. Re-charging battery, if necessary, following the manufacturer's guidelines 2.8. Inspecting electric motor and inverter system, their cooling system, electrical connections and wiring therein, and motor brushes for wear 2.9. Performing hybrid system diagnostics using appropriate tools and software, 2.10. Testing and inspecting power electronics system, wiring and electrical connections, transmission system, regenerative braking functionality and hybrid system control modules
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Personal protective equipment (PPE) 4.3. Tools and equipment 4.4. Measuring and testing instruments

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood hybrid system components 1.2 diagnosed hybrid system faults 1.3 serviced hybrid batteries 1.4 maintained electric motor and inverter systems 1.5 troubleshooted hybrid system issues
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Workshop/Lab Facility Standard

Course Name:	Auto Mechanics
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer	02
2.	Projector	01
3.	Internet connectivity	01
4.	Carburettor engine	02
5.	EFI engine	04
6.	VVTi engine	02
7.	Hybrid engine	01
8.	Manual gear system	02
9.	Power gear system	02
10.	Power transmission system	01
11.	Tire changer machine	01
12.	Wheel balancing machine	01
13.	Hydraulic lift	01
14.	Gasoline injector cleaner	01
15.	Exhaust gas analyser	01
16.	Engine scanner	01
17.	On board diagnostic (OBD) scanner	01
18.	Cut-way engine model (Disel and Petrol)	01
19.	Trainer board of auto electric system	01
20.	Trainer board of brake system	01
21.	Trainer board of transmission system	01
22.	High pressure pump (Disel)	01
23.	Complete tool box (with tools)	25

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Auto Mechanics, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.